Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Bluestock Mutual Fund Analytics Capstone Project

Table Of Contents:

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Executive Summary :

The Bluestock Mutual Fund Analytics Capstone Project was developed to analyze mutual fund performance, investor behavior, and industry trends using data analytics and business intelligence techniques. The project integrates multiple datasets, including scheme performance, NAV history, investor transactions, portfolio holdings, and market benchmarks, to provide a comprehensive view of the mutual fund ecosystem. An ETL pipeline was implemented to clean, transform, and organize the data for analysis and reporting.

Advanced financial metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were used to evaluate fund performance and risk. Investor transaction data was analyzed to understand investment patterns, SIP continuity, and cohort behavior, while portfolio holdings were examined to measure diversification and concentration across sectors.

The final deliverables include an interactive Power BI dashboard with four analytical pages and a set of advanced analytics models developed in Python. The project successfully demonstrates how data engineering, financial analytics, and visualization tools can be combined to generate actionable insights for investors, fund managers, and financial analysts.

Introduction

Mutual funds have become one of the most popular investment vehicles for individuals seeking diversified exposure to financial markets. With the rapid growth of Assets Under Management (AUM), increasing SIP participation, and a wide variety of investment schemes, analyzing mutual fund performance has become essential for both investors and fund managers. Data-driven insights can help identify performance trends, risk characteristics, investor behavior, and market opportunities.

The objective of this project is to build a comprehensive Mutual Fund Analytics platform that combines data engineering, financial analysis, and business intelligence techniques. By integrating multiple datasets and applying analytical models, the project aims to provide meaningful insights into industry growth, fund performance, investor transactions, and portfolio composition.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

The project leverages Python, SQL, SQLite, Jupyter Notebook, and Power BI to develop an end-to-end analytics solution. The final outcome includes interactive dashboards, advanced financial metrics, and investor-focused analytics that support informed decision-making in the mutual fund industry.

Objectives

The primary objective of this project is to develop an end-to-end Mutual Fund Analytics solution that enables comprehensive analysis of mutual fund performance, investor behavior, and industry trends. The project aims to transform raw financial data into actionable insights through data engineering, analytics, and visualization techniques.

The key objectives of the project are:

- Design and implement an ETL pipeline to clean and integrate multiple mutual fund datasets.
- Analyze fund performance using financial metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Study investor behavior through transaction analysis, cohort analysis, and SIP continuity assessment.
- Develop advanced analytics models including VaR, CVaR, Rolling Sharpe Ratio, Fund Recommendation System, and Sector Concentration Analysis.
- Create an interactive Power BI dashboard to visualize industry trends, fund performance, investor analytics, and SIP market insights.
- Generate business recommendations and insights to support informed investment decisions.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Data Sources

The Bluestock Mutual Fund Analytics project utilizes multiple datasets covering mutual fund schemes, historical performance, investor transactions, portfolio composition, and industry trends. These datasets were cleaned, validated, and integrated through the ETL pipeline to support analytics and dashboard development.

1. Fund Master Dataset (01_fund_master_cleaned)

This dataset contains the master information for mutual fund schemes, including AMFI code, scheme name, fund house, category, and plan type. It serves as the primary reference dataset for linking all other mutual fund data.

2. NAV History Dataset (02_nav_history_cleaned)

The NAV History dataset contains daily Net Asset Value (NAV) records for mutual fund schemes from 2022 to 2025. This dataset was used to calculate daily returns, rolling Sharpe ratios, Value at Risk (VaR), and Conditional Value at Risk (CVaR).

3. AUM by Fund House Dataset (03_aum_by_fund_house_cleaned)

This dataset contains Assets Under Management (AUM) values for different fund houses. It was used to analyze industry size, compare fund houses, and identify leading asset management companies.

4. Monthly SIP Inflows Dataset (04_monthly_sip_inflows_cleaned)

This dataset contains monthly SIP inflow information and was used to study investment trends, SIP growth patterns, and overall retail participation in mutual funds.

5. Category Inflows Dataset (05_category_inflows_cleaned)

The category inflows dataset contains category-wise investment inflows across different mutual fund categories such as Large Cap, Mid Cap, Flexi Cap, and Liquid Funds. It supports category-level trend analysis and inflow comparisons.

6. Industry Folio Count Dataset (06_industry_folio_count_cleaned)

This dataset contains industry-wide folio statistics and was used to measure investor participation and growth in mutual fund investments over time.

7. Scheme Performance Dataset (07_scheme_performance_cleaned)

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

This dataset contains key performance indicators for each mutual fund scheme, including returns, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, AUM, expense ratio, and risk grade. It forms the basis of the performance analysis conducted in the project.

8. Investor Transactions Dataset (08_investor_transactions_cleaned)

The investor transactions dataset includes SIP, lump-sum, and redemption transactions along with investor demographics such as age group, state, city tier, gender, and income category. It was used to analyze investor behavior, transaction patterns, cohort trends, and SIP continuity.

9. Portfolio Holdings Dataset (09_portfolio_holdings_cleaned)

This dataset contains stock-level holdings and sector allocations for mutual fund schemes. It was used to evaluate portfolio diversification and concentration using the Herfindahl-Hirschman Index (HHI).

10. Benchmark Indices Dataset (10_benchmark_indices_cleaned)

The benchmark dataset contains market index values used to compare mutual fund performance against broader market movements. It supports benchmark analysis and NAV comparison visualizations.

Together, these datasets provide a comprehensive view of the mutual fund ecosystem and enable detailed analysis of fund performance, investor behavior, portfolio composition, and market trends.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

ETL Design

The Bluestock Mutual Fund Analytics project follows a structured ETL (Extract, Transform, Load) framework to ensure data quality, consistency, and reliability throughout the analytics lifecycle. The ETL pipeline was designed to integrate multiple mutual fund datasets from different sources and prepare them for analysis, reporting, and dashboard development.

Extract Phase

The extraction phase involved collecting raw data from multiple CSV datasets related to mutual fund schemes, NAV history, SIP inflows, category inflows, industry folio counts, investor transactions, portfolio holdings, benchmark indices, and fund house AUM statistics. These datasets formed the foundation of the analytical workflow and were stored in a dedicated raw data repository before processing.

Transform Phase

The transformation stage focused on cleaning, validating, and standardizing the extracted datasets. Duplicate records were identified and removed using Pandas functions to ensure data accuracy. Missing values were detected using data quality checks and were either filled or removed based on business relevance and analytical requirements. Critical columns required for financial calculations and dashboard reporting were verified to contain complete and valid information.

Date fields across all datasets were standardized using the datetime format to maintain consistency during time-series analysis. Examples include NAV history dates, investor transaction dates, SIP inflow records, benchmark index dates, and industry statistics. Column names were also standardized by converting them to lowercase and replacing spaces with underscores, resulting in consistent naming conventions such as fund_house, amfi_code, and transaction_date.

In addition to data cleaning, several derived analytical datasets were created. These include alpha_beta.csv for benchmark-based risk metrics, cagr_comparison.csv for growth analysis, fund_scorecard.csv for composite fund ranking, and performance_anomalies.csv for identifying unusual fund behavior. Advanced metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, CVaR, Rolling Sharpe Ratio, and HHI concentration scores were generated during the transformation and analytics stages.

Load Phase

After transformation, the cleaned datasets were stored in both processed CSV files and a SQLite database. Processed CSV files served as the primary source for Jupyter Notebook analysis and Power BI reporting, while SQLite provided structured storage and supported ETL validation activities. The final datasets were used to perform Exploratory Data Analysis (EDA), Performance Analytics, Advanced Analytics, and Dashboard Development.

ETL Pipeline Flow

Raw CSV Files
→ Data Cleaning & Validation
→ Processed CSV Files
→ SQLite Database
→ EDA & Performance Analytics
→ Derived Datasets
→ Power BI Dashboard

This ETL pipeline ensured that all analytical outputs and dashboard visualizations were built on reliable, standardized, and high-quality data, enabling accurate financial analysis and meaningful business insights.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Exploratory Data Analysis (EDA) Findings

Exploratory Data Analysis (EDA) was conducted to understand mutual fund performance trends, investor demographics, industry growth, and portfolio allocation patterns. Multiple visualizations were created to identify relationships, distributions, and emerging trends within the datasets.

NAV and Fund Performance Trends

The Daily NAV Trend analysis revealed that most mutual fund schemes experienced a steady upward trajectory between 2022 and 2026, despite periodic market corrections and short-term volatility. This indicates overall growth in fund valuations over the study period. Among all analyzed schemes, Fund 120844 recorded the highest NAV value, significantly outperforming many other funds in the dataset.

The Top 10 Funds by Latest NAV analysis highlighted substantial differences in fund valuations and demonstrated the presence of market leaders with consistently strong performance.

AUM and Industry Growth

The AUM Growth Analysis showed that SBI Mutual Fund maintained the largest Assets Under Management (AUM) among the analyzed fund houses, reinforcing its position as a market leader. The steady increase in industry AUM reflects growing investor participation and confidence in mutual fund investments.

SIP Trend Analysis

Monthly SIP inflow analysis demonstrated a strong upward trend throughout the study period. SIP inflows reached a record high of ₹31,002 Crore in December 2025, reflecting increased investor confidence and growing awareness of systematic investment strategies.

The SIP Year-over-Year Growth Analysis showed accelerated participation during 2024 and 2025, suggesting that retail investors increasingly preferred disciplined and long-term investment approaches.

Category and Sector Analysis

The Category Inflow Heatmap revealed that Liquid Funds consistently attracted the highest net inflows across multiple periods. This indicates a

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

preference among investors for liquidity, capital preservation, and relatively lower investment risk.

Sector Allocation Analysis showed that Financial Services and Information Technology received the largest portfolio allocations across equity-oriented mutual funds. Consumer-related sectors also maintained significant representation within fund portfolios.

Investor Demographics and Geographic Distribution

Investor demographic analysis showed that the 25–40 age group represented the largest share of investors, making it the dominant segment within the mutual fund ecosystem. This indicates strong participation from working professionals and early-stage wealth creators.

Geographic analysis revealed that investment participation expanded across multiple states rather than being concentrated in a limited number of regions. The State-wise Distribution analysis indicated broad investor engagement across the country.

Equity vs Debt Participation

The Equity vs Debt Distribution analysis demonstrated that equity-oriented funds dominated investor participation compared to debt-oriented funds. This reflects investors' preference for long-term wealth creation and higher return potential despite increased market risk.

Correlation Analysis

The NAV Return Correlation Matrix identified strong correlations among several large-cap schemes, indicating that these funds often moved in similar directions due to common market influences. However, the analysis also

revealed diversification opportunities across categories and fund types, which can help investors manage portfolio risk effectively.

Overall, the EDA phase provided valuable insights into fund performance, investor behavior, industry growth, and portfolio allocation patterns, forming the foundation for subsequent performance and advanced analytics.

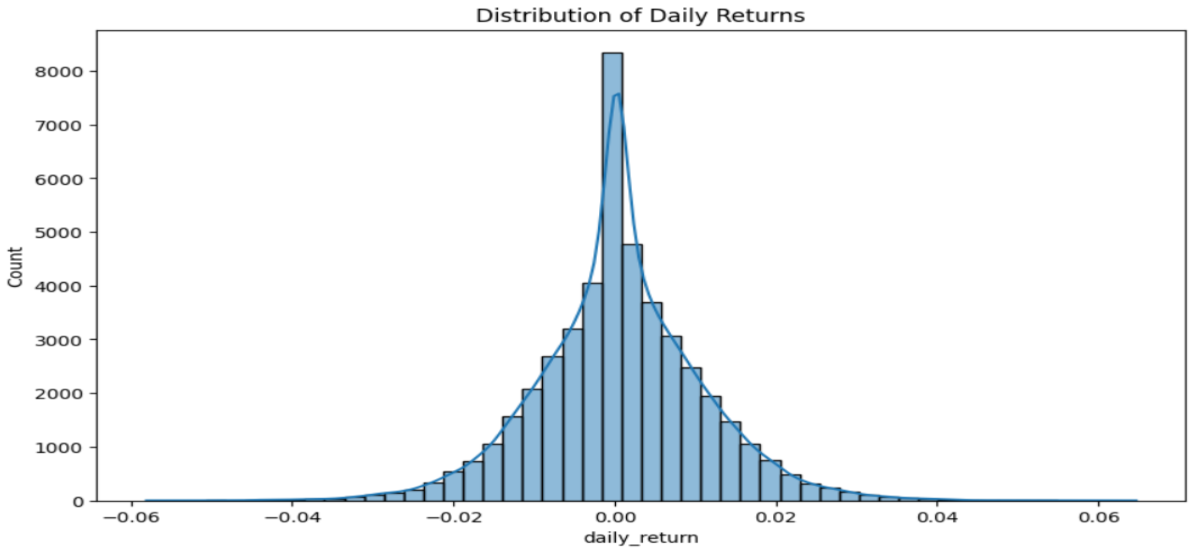
Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Performance Analysis

Performance analysis was conducted to evaluate mutual fund schemes using a combination of return-based, risk-based, and benchmark-relative performance metrics. The objective was to identify funds that delivered strong returns while effectively managing risk and maintaining consistency over time.

Return Analysis

Daily return analysis showed that return distributions were generally centered near zero with realistic volatility patterns. This indicates that the datasets captured normal market fluctuations and provided a reliable foundation for further performance evaluation. CAGR (Compound Annual Growth Rate) analysis revealed that several schemes generated strong three-year and full-period growth rates, demonstrating their ability to create long-term wealth for investors.



Risk-Adjusted Performance

Risk-adjusted performance was evaluated using Sharpe Ratio and Sortino Ratio. Sharpe Ratio measures excess return generated per unit of total risk, while Sortino Ratio focuses specifically on downside risk.

The analysis showed that Funds 100033 and 101206 achieved Sharpe Ratios greater than 1, indicating strong risk-adjusted performance relative to other schemes. Sortino Ratio results further confirmed that top-performing funds were effective in limiting downside volatility while maintaining attractive returns.

These findings highlight the importance of evaluating risk-adjusted metrics rather than relying solely on absolute returns when comparing mutual funds.

Alpha and Beta Analysis

Alpha and Beta were calculated using the NIFTY100 benchmark to measure benchmark-relative performance and market sensitivity.

Several schemes generated positive Alpha values, indicating that they outperformed the benchmark after adjusting for market risk. Positive Alpha suggests that fund managers were able to add value beyond what would be expected from general market movements.

Beta values remained relatively close to zero for many schemes, indicating weak correlation with benchmark movements within the available dataset. This suggests that several funds exhibited unique return characteristics that were not entirely driven by benchmark performance.

Maximum Drawdown Analysis

Maximum Drawdown was used to evaluate downside risk by measuring the largest decline from a historical peak to a subsequent trough.

The analysis revealed significant variation across schemes, with drawdowns ranging from approximately -0.1% to -52.6%. Funds with smaller drawdowns demonstrated greater resilience during adverse market conditions, while funds with larger drawdowns experienced higher downside exposure.

This metric provides valuable insight into the potential losses investors may experience during periods of market stress.

Fund Scorecard Evaluation

A composite fund scorecard was developed to provide a holistic evaluation framework by combining multiple performance indicators, including returns, Sharpe Ratio, Alpha, Expense Ratio, and Maximum Drawdown.

Benchmark Comparison and Tracking Error

Benchmark comparison was performed to assess how closely mutual fund performance aligned with market indices. Tracking Error analysis revealed meaningful differences among schemes in their ability to replicate benchmark performance.

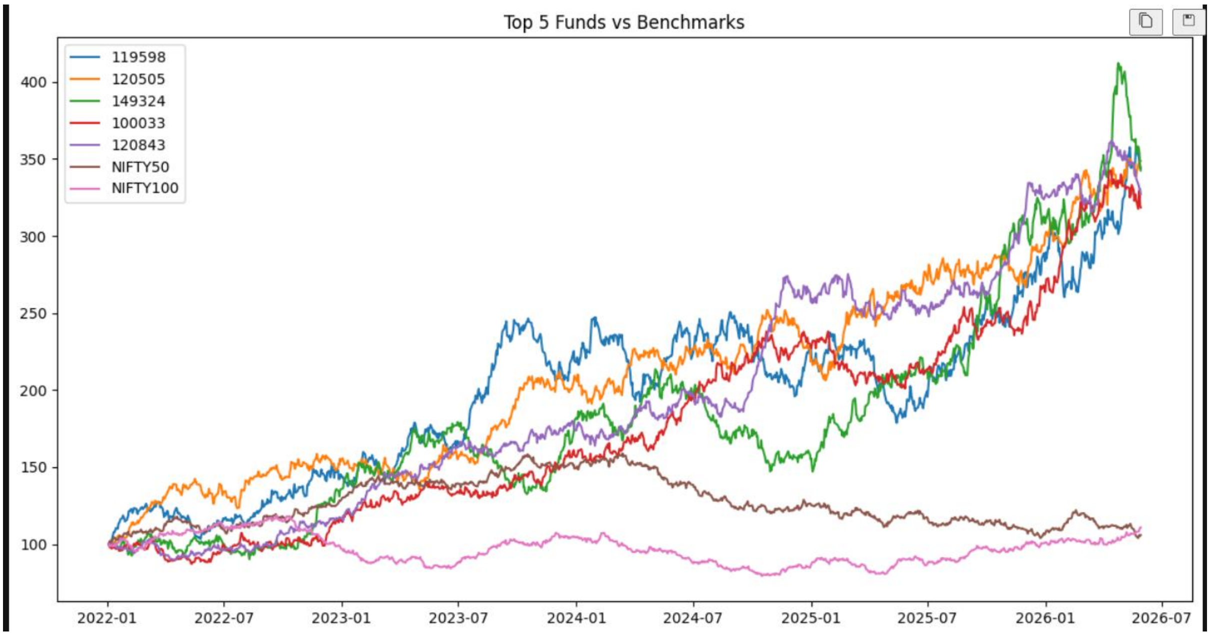
Some funds closely tracked benchmark movements, while others displayed larger deviations due to active management strategies and portfolio positioning decisions.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Key Findings

The performance analysis demonstrates that evaluating mutual funds requires a combination of return, risk, cost, and benchmark-relative metrics. While high returns remain important, metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Maximum Drawdown, and Tracking Error provide deeper insight into overall fund quality and investment suitability.

The results show that funds with strong risk-adjusted returns and controlled downside risk generally provide a more balanced investment profile than funds evaluated solely on return performance.



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Dashboard Overview

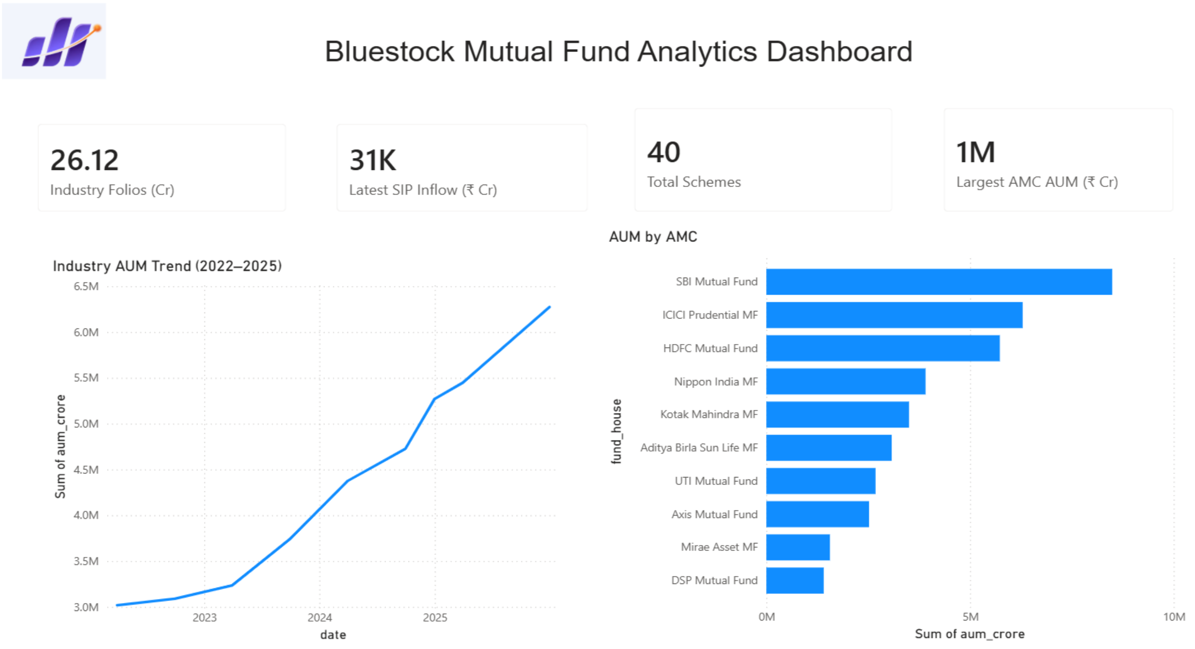


Figure 1: Industry Overview Dashboard

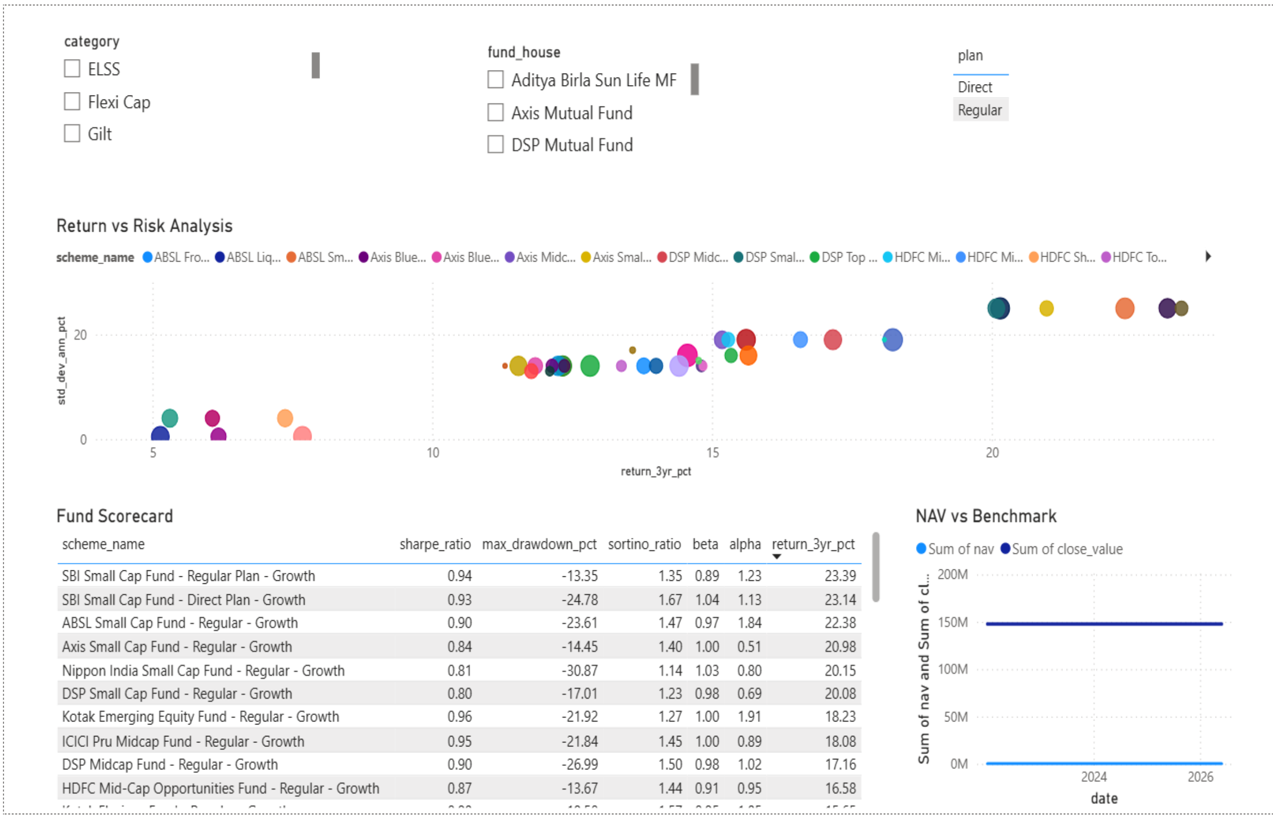
Dashboard Page 1 – Industry Overview

The Industry Overview dashboard provides a high-level view of the mutual fund industry's growth and key performance indicators. It highlights important metrics such as Industry Folios (26.12 Crore), Monthly SIP Inflows (₹31K Crore), Total Schemes (40), and the Largest AMC by Assets Under Management (AUM). These KPIs offer a quick summary of investor participation and overall market size.

The page also includes an Industry AUM Trend chart showing consistent growth in AUM from 2022 to 2025, indicating increasing investor confidence in mutual funds. The AUM by AMC visualization compares major fund houses and shows that SBI Mutual Fund manages the highest AUM among the selected AMCs, followed by ICICI Prudential and HDFC Mutual Fund. Together, these visualizations provide a clear overview of industry growth and market leadership.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Dashboard Page 2 – Fund Performance



scheme_name	sharpe_ratio	max_drawdown_pct	sortino_ratio	beta	alpha	return_3yr_pct
SBI Small Cap Fund - Regular Plan - Growth	0.94	-13.35	1.35	0.89	1.23	23.39
SBI Small Cap Fund - Direct Plan - Growth	0.93	-24.78	1.67	1.04	1.13	23.14
ABSL Small Cap Fund - Regular - Growth	0.90	-23.61	1.47	0.97	1.84	22.38
Axis Small Cap Fund - Regular - Growth	0.84	-14.45	1.40	1.00	0.51	20.98
Nippon India Small Cap Fund - Regular - Growth	0.81	-30.87	1.14	1.03	0.80	20.15
DSP Small Cap Fund - Regular - Growth	0.80	-17.01	1.23	0.98	0.69	20.08
Kotak Emerging Equity Fund - Regular - Growth	0.96	-21.92	1.27	1.00	1.91	18.23
ICICI Pru Midcap Fund - Regular - Growth	0.95	-21.84	1.45	1.00	0.89	18.08
DSP Midcap Fund - Regular - Growth	0.90	-26.99	1.50	0.98	1.02	17.16
HDFC Mid-Cap Opportunities Fund - Regular - Growth	0.87	-13.67	1.44	0.91	0.95	16.58

Figure 2: Fund Performance Dashboard

The Fund Performance dashboard focuses on evaluating mutual fund schemes using return, risk, and benchmark-related metrics. Interactive filters allow users to analyze funds by category, fund house, and plan type, enabling detailed comparison across different investment options.

The Return vs Risk Analysis scatter plot visualizes the relationship between 3-year returns and annualized volatility, helping identify funds that deliver higher returns for a given level of risk. The Fund Scorecard provides key performance indicators such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and returns, allowing investors to compare schemes comprehensively. The NAV vs Benchmark chart further enables performance comparison against market indices, helping assess whether funds are outperforming or underperforming benchmark expectations.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Dashboard Page 3 – Investor Analytics

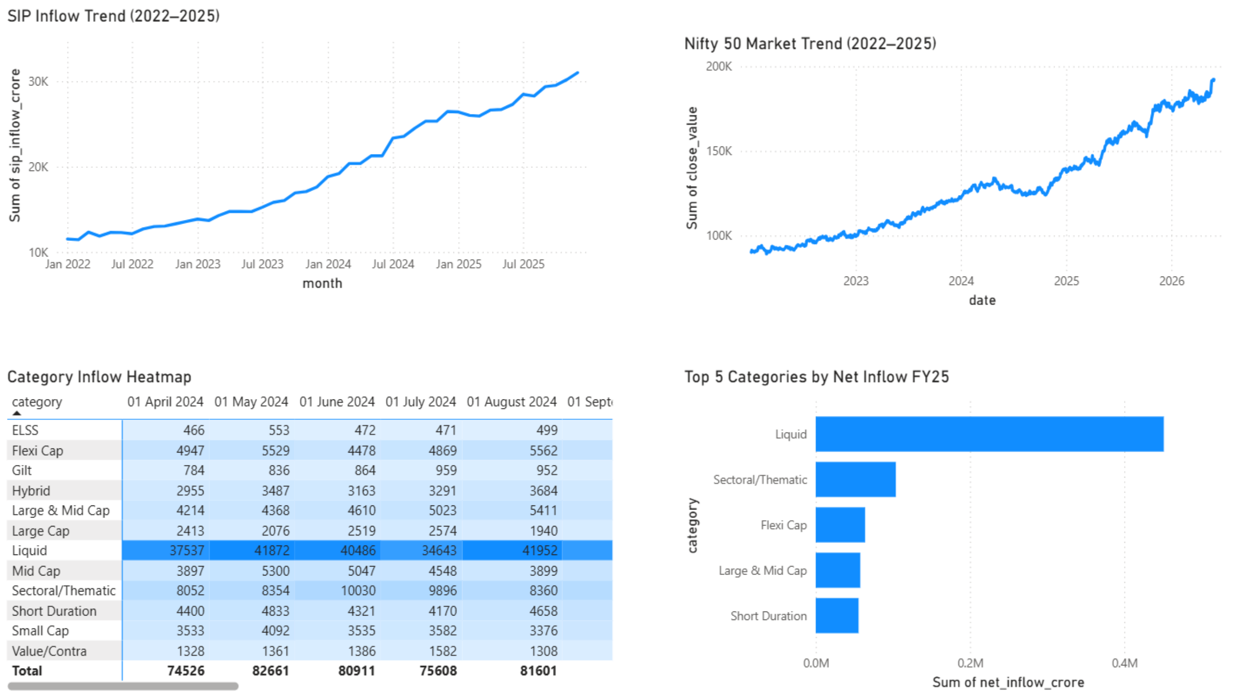
Figure 3: Investor Analytics Dashboard

The Investor Analytics dashboard provides insights into investor behavior, transaction patterns, and demographic trends. Interactive filters allow users to analyze investment activity based on age group, city tier, and state, helping identify key investor segments and regional participation patterns.

The dashboard highlights transaction amounts across different states, monthly transaction volume trends, investment type distribution, and average investment amounts by age group. The analysis shows that investment activity is distributed across multiple states, while the 26–35 age group contributes a significant share of investments. The Investment Type Distribution chart indicates that lump-sum investments account for the largest proportion of transactions, followed by redemptions and SIP investments. Together, these visualizations help understand investor preferences, participation levels, and transaction behavior across different demographic groups.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Dashboard Page 4 – SIP & Market Trends



category	01 April 2024	01 May 2024	01 June 2024	01 July 2024	01 August 2024	01 Sept
ELSS	466	553	472	471	499	
Flexi Cap	4947	5529	4478	4869	5562	
Gilt	784	836	864	959	952	
Hybrid	2955	3487	3163	3291	3684	
Large & Mid Cap	4214	4368	4610	5023	5411	
Large Cap	2413	2076	2519	2574	1940	
Liquid	37537	41872	40486	34643	41952	
Mid Cap	3897	5300	5047	4548	3899	
Sectoral/Thematic	8052	8354	10030	9896	8360	
Short Duration	4400	4833	4321	4170	4658	
Small Cap	3533	4092	3535	3582	3376	
Value/Contra	1328	1361	1386	1582	1308	
Total	**74526**	**82661**	**80911**	**75608**	**81601**	

Figure 4: SIP & Market Trends Dashboard

The SIP & Market Trends dashboard focuses on investment inflows and overall market performance. The SIP Inflow Trend chart shows a steady increase in monthly SIP contributions from 2022 to 2025, reaching approximately ₹31,000 Crore by the end of the analysis period. This growth reflects increasing retail investor participation and confidence in long-term systematic investing.

The dashboard also compares SIP growth with the Nifty 50 market trend, illustrating how investor inflows evolved alongside market performance. The Category Inflow Heatmap highlights investment activity across mutual fund categories, while the Top 5 Categories by Net Inflow visualization shows that Liquid Funds attracted the highest inflows during FY25, followed by Sectoral/Thematic and Flexi Cap funds. These insights help identify investor preferences and emerging trends within the mutual fund industry.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Advanced Analytics

To extend the scope of traditional mutual fund analysis, several advanced analytical techniques were implemented to evaluate risk, investor behavior, portfolio concentration, and fund suitability. These analyses provide deeper insights beyond standard return-based metrics and support more informed investment decision-making.

Historical VaR and CVaR Analysis

Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated using daily return distributions for all mutual fund schemes. VaR estimates the maximum expected loss under normal market conditions at a 95% confidence level, while CVaR measures the average loss beyond the VaR threshold. The analysis revealed significant differences in downside risk across schemes, helping identify funds that may be more vulnerable during adverse market conditions.

Rolling Sharpe Ratio Analysis

A 90-day Rolling Sharpe Ratio was computed to evaluate how risk-adjusted performance changed over time. Unlike a single Sharpe Ratio value, rolling analysis provides a dynamic view of fund performance across different market environments. The results showed that several funds maintained consistently positive risk-adjusted returns, while others experienced fluctuations during periods of market volatility.

Investor Cohort Analysis

Investor cohorts were created based on the year of their first transaction. The analysis compared average SIP amounts, total investments, and preferred mutual fund schemes across different investor groups. The findings indicated that newer investor cohorts contributed significantly to total investments, reflecting growing participation in mutual fund investing and increasing awareness of systematic investment strategies.

SIP Continuity Analysis

SIP continuity was assessed by analyzing the average gap between SIP transactions for investors with at least six SIP contributions. Investors with an average gap exceeding 35 days were classified as at-risk investors. This

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

analysis helps identify potential declines in investment discipline and can support proactive investor engagement strategies.

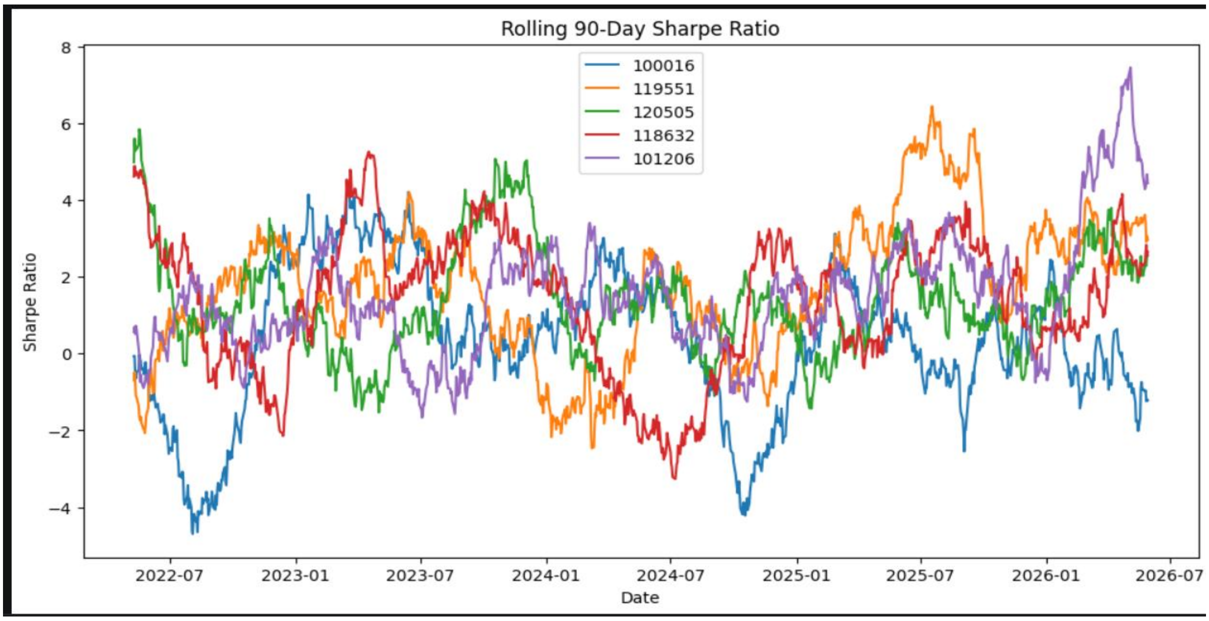
Fund Recommendation System

A rule-based fund recommendation system was developed using risk appetite categories such as Low, Moderate, and High. Funds were ranked according to Sharpe Ratio within each risk category, and the top-performing schemes were recommended. This approach demonstrates how analytics can be used to support personalized investment decision-making based on investor risk tolerance.

Sector Concentration Analysis

Portfolio diversification was evaluated using the Herfindahl-Hirschman Index (HHI), which measures sector concentration within mutual fund portfolios. Higher HHI values indicate concentrated portfolios, while lower values indicate greater diversification. The analysis revealed noticeable differences in sector allocation strategies among funds, helping investors understand concentration risk and diversification characteristics.

Overall, the advanced analytics component enhanced the project's analytical depth by combining financial risk measurement, investor behavior analysis, recommendation techniques, and portfolio concentration assessment. These insights complement the dashboard findings and provide a more comprehensive understanding of mutual fund performance and investor activity.



Limitations

While the project provides comprehensive insights into mutual fund performance and investor behavior, certain limitations should be acknowledged. The analysis is based on the datasets provided for the capstone project and may not fully represent the entire mutual fund industry. Real-time market data and live NAV updates were not integrated, limiting the analysis to historical observations.

Additionally, the fund recommendation system is based on historical performance metrics and risk classifications, which may not accurately predict future returns. Market conditions, economic events, and investor sentiment can significantly influence future fund performance. Therefore, the insights generated should be considered as analytical guidance rather than direct investment advice.

Recommendations

Based on the analysis conducted during the project, several recommendations can be made to enhance future mutual fund analytics and decision-making processes. Integrating real-time NAV feeds and market data would improve the accuracy and relevance of performance monitoring. Automated data refresh mechanisms can further streamline the ETL process and ensure dashboards remain up to date.

Advanced machine learning models can be incorporated to predict fund performance, identify investment opportunities, and improve fund recommendation accuracy. Additional investor segmentation techniques may also be implemented to better understand behavioral patterns across different demographic groups. Finally, deploying the dashboard through Power BI Service would enable wider accessibility, real-time collaboration, and more effective business reporting.

Name: Viyyapu Vinay
Email id: viyyapuvinay939@gmail.com

Conclusion

The Bluestock Mutual Fund Analytics Capstone Project successfully demonstrates an end-to-end analytics workflow that combines data engineering, financial analysis, advanced analytics, and business intelligence reporting. Through the implementation of a structured ETL pipeline, multiple datasets were cleaned, integrated, and transformed into meaningful insights that support the evaluation of mutual fund performance and investor behavior.

The project delivered interactive Power BI dashboards, advanced risk analytics, investor cohort analysis, SIP continuity assessment, portfolio concentration analysis, and a fund recommendation system. The results highlight the value of combining financial metrics, investor analytics, and visualization tools to support data-driven decision-making. Overall, the project showcases practical applications of Python, SQL, Power BI, and financial analytics in solving real-world problems within the mutual fund industry.

The project successfully achieved all planned objectives and demonstrates the practical application of data engineering, financial analytics, and business intelligence techniques in the mutual fund industry.